GLOBAL SOLAR ATLAS
BY WORLD BANK GROUP

Dongola

19.164627°, 030.478821°
unnamed road, Dongola, Northern State, Sudan
Time zone: UTC+02, Africa/Khartoum [CAT]

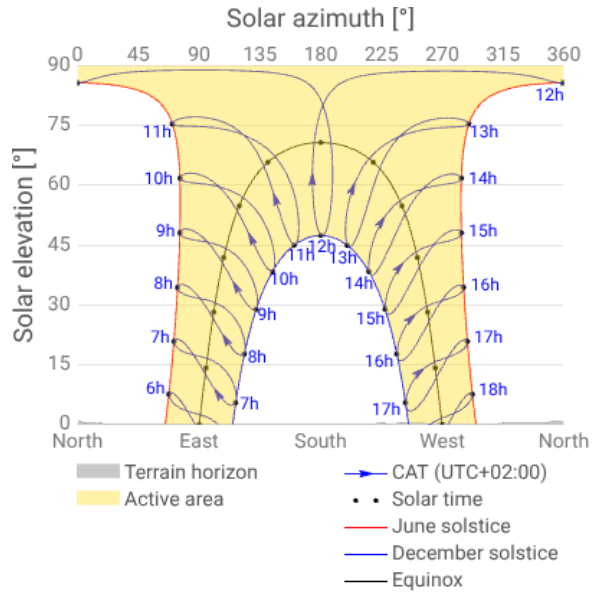
Report generated: 4 Nov 2026

SITE INFO

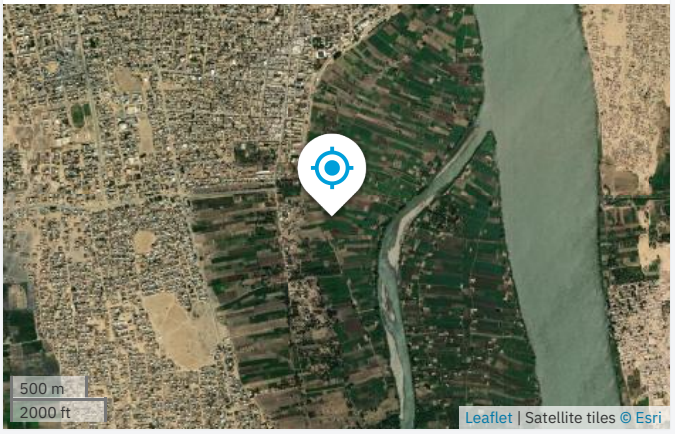
Map data

		Per year	
Specific photovoltaic power output	PVOUT specific	**1883.8**	kWh/kWp
Direct normal irradiation	DNI	**2029.0**	kWh/m²
Global horizontal irradiation	GHI	**2335.6**	kWh/m²
Diffuse horizontal irradiation	DIF	**925.5**	kWh/m²
Global tilted irradiation at optimum angle	GTI opta	**2474.7**	kWh/m²
Optimum tilt of PV modules	OPTA	**22 / 180**	°
Air temperature	TEMP	**28.8**	°C
Terrain elevation	ELE	**226**	m

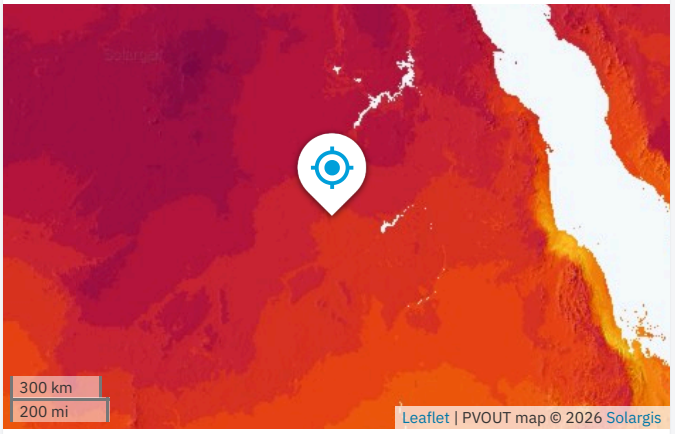
Horizon and sunpath



Map



PVOUT map



PVOUT: Long-term average of annual totals of PV power potential

kWh/kWp
600 800 1000 1200 1400 1600 1800 2000 2200 2400

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PV ELECTRICITY AND SOLAR RADIATION

Annual averages

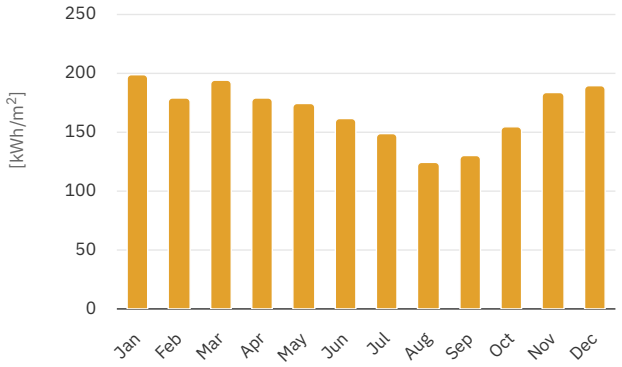
Direct normal irradiation

2012.4

kWh/m^2 per year

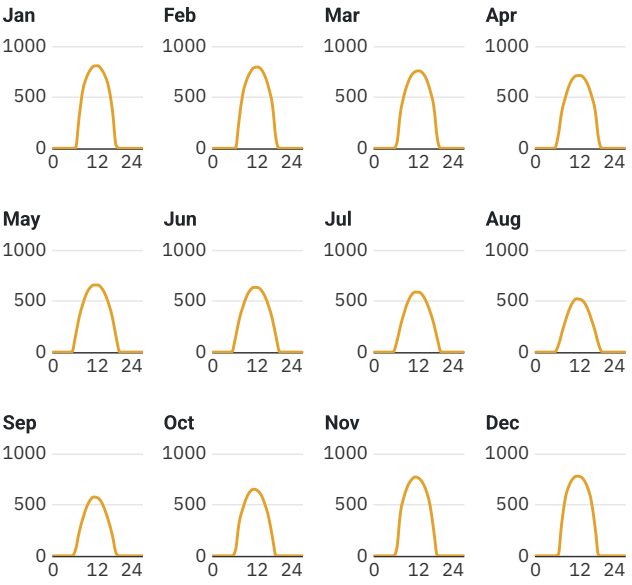
Monthly averages

Direct normal irradiation



Average hourly profiles

Direct normal irradiation [Wh/m^2]



UTC+02

Average hourly profiles

Direct normal irradiation [Wh/m^2]

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0 - 1												
1 - 2												
2 - 3												
3 - 4												
4 - 5												
5 - 6												
6 - 7	4	1	47	116	153	136	99	57	57	52	70	8
7 - 8	306	277	346	357	340	314	253	185	231	319	420	323
8 - 9	570	531	521	500	470	439	383	307	365	461	581	570
9 - 10	693	663	639	614	573	542	490	414	476	566	685	689
10 - 11	767	743	715	683	632	605	559	489	546	630	744	754
11 - 12	803	787	750	710	656	632	586	520	575	651	771	779
12 - 13	807	794	757	712	655	631	585	515	566	637	757	774
13 - 14	773	771	736	686	628	604	555	487	526	591	714	739
14 - 15	706	703	676	624	566	542	490	420	442	500	631	664
15 - 16	589	589	566	517	467	445	394	318	330	374	492	530
16 - 17	362	420	408	370	332	319	276	205	193	178	242	275
17 - 18	25	83	85	79	141	144	123	67	24	2	1	2
18 - 19						1	1					
19 - 20												
20 - 21												
21 - 22												
22 - 23												
23 - 24												
Sum	6,407	6,361	6,245	5,968	5,614	5,353	4,794	3,985	4,332	4,962	6,108	6,106

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GLOSSARY

Acronym	Full name	Unit	Type of use
DIF	Diffuse horizontal irradiation	kWh/m², MJ/m²	Average yearly, monthly or daily sum of diffuse horizontal irradiation (© 2025 Solargis)
DNI	Direct normal irradiation	kWh/m², MJ/m²	Average yearly, monthly or daily sum of direct normal irradiation (© 2025 Solargis)
ELE	Terrain elevation	m, ft	Elevation of terrain surface above/below sea level, processed and integrated from SRTM-3 data and related data products (SRTM v4.1 © 2004 - 2025, CGIAR-CSI)
GHI	Global horizontal irradiation	kWh/m², MJ/m²	Average annual, monthly or daily sum of global horizontal irradiation (© 2025 Solargis)
GTI	Global tilted irradiation	kWh/m², MJ/m²	Average annual, monthly or daily sum of global tilted irradiation (© 2025 Solargis)
GTI_opta	Global tilted irradiation at optimum angle	kWh/m², MJ/m²	Average annual, monthly or daily sum of global tilted irradiation for PV modules fix-mounted at optimum angle (© 2025 Solargis)
OPTA	Optimum tilt of PV modules	°	Optimum tilt of fix-mounted PV modules facing towards Equator set for maximizing GTI input (© 2025 Solargis)
PVOUT_total	Total photovoltaic power output	kWh, MWh, GWh	Yearly and monthly average values of photovoltaic electricity (AC) delivered by the total installed capacity of a PV system (© 2025 Solargis)
PVOUT_specific	Specific photovoltaic power output	kWh/kWp	Yearly and monthly average values of photovoltaic electricity (AC) delivered by a PV system and normalized to 1 kWp of installed capacity (© 2025 Solargis)
TEMP	Air temperature	°C, °F	Average yearly, monthly and daily air temperature at 2 m above ground. Calculated from outputs of ERA5 model (© 2025 ECMWF, post-processed by Solargis)

ABOUT

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